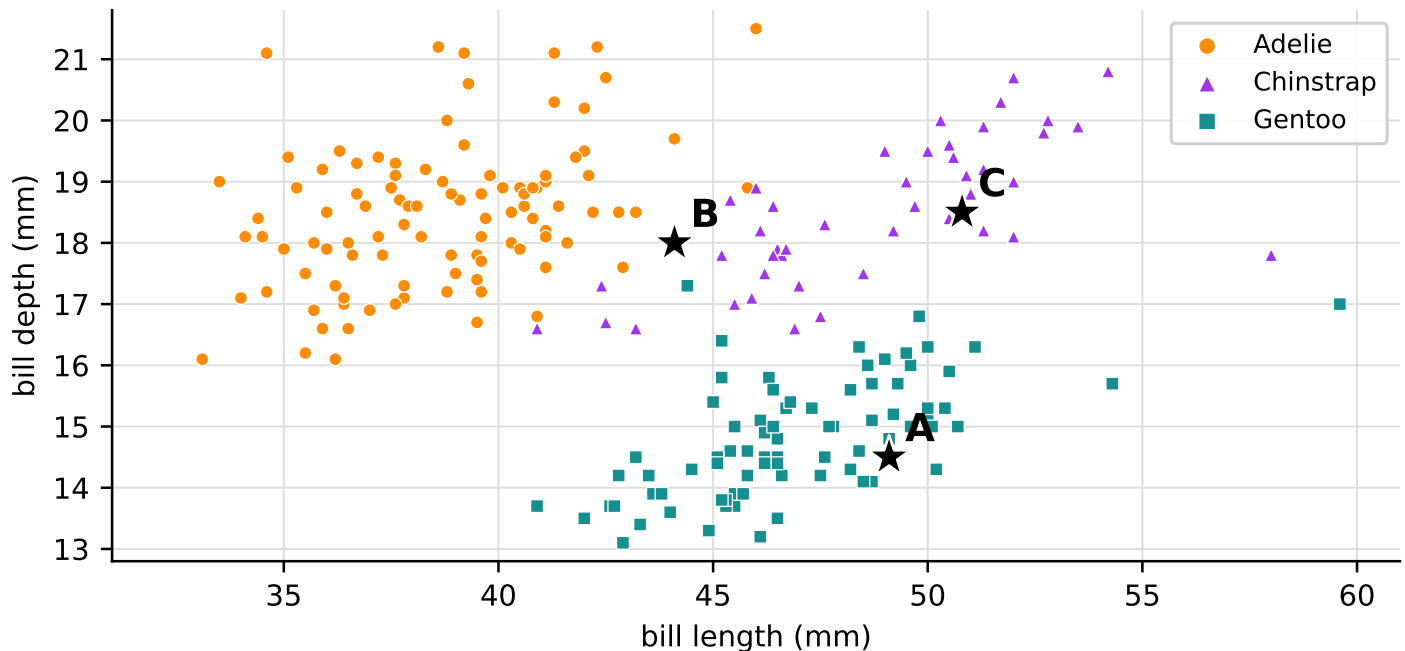


DATA 202 — k-Nearest Neighbors · Week 6 Monday

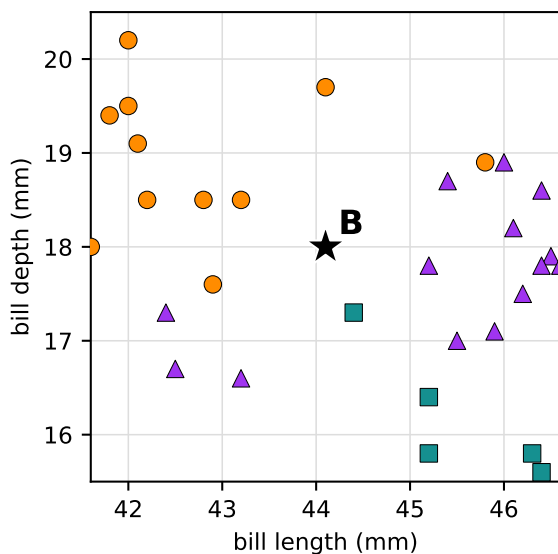
Draw right on the plots. They show the same penguins as today's slides: the 223 measured in 2007–2008 (circles = Adelie, triangles = Chinstrap, squares = Gentoo) and three new ones from 2009 (★). Your professor will tell you when to do each part.

1 · Which species?



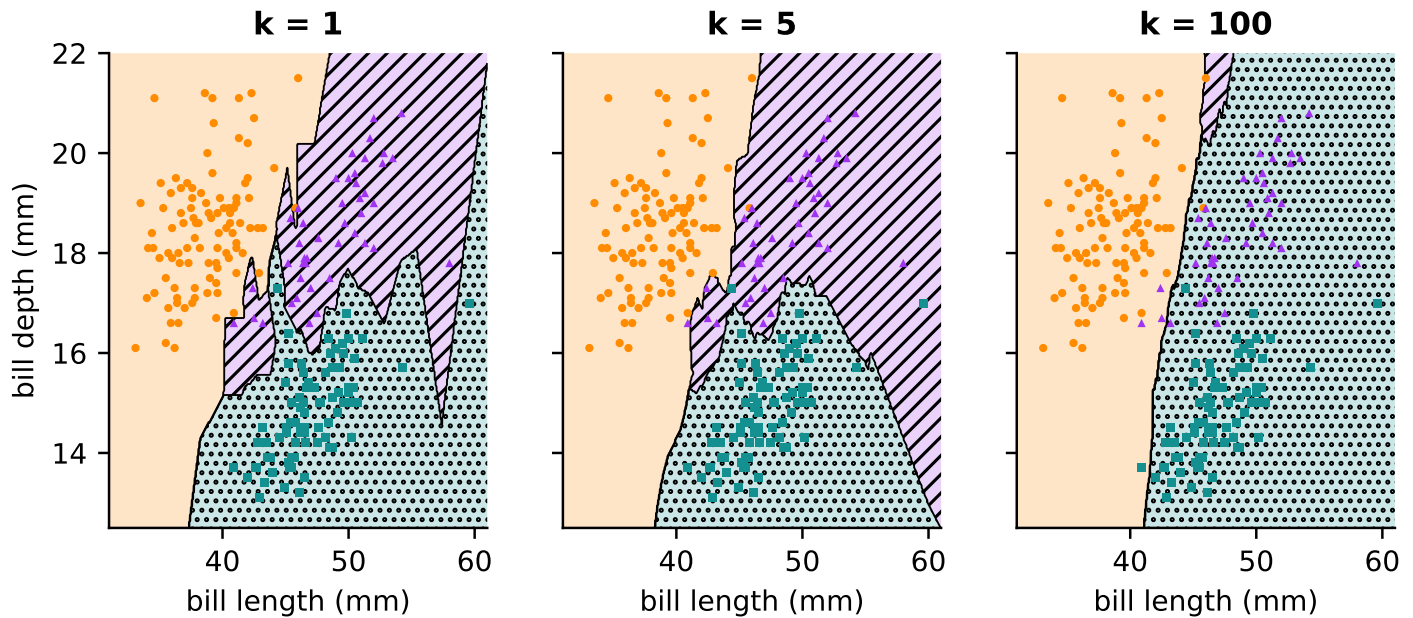
Your guess: A _____ B _____ C _____

2 · Ask the neighbors

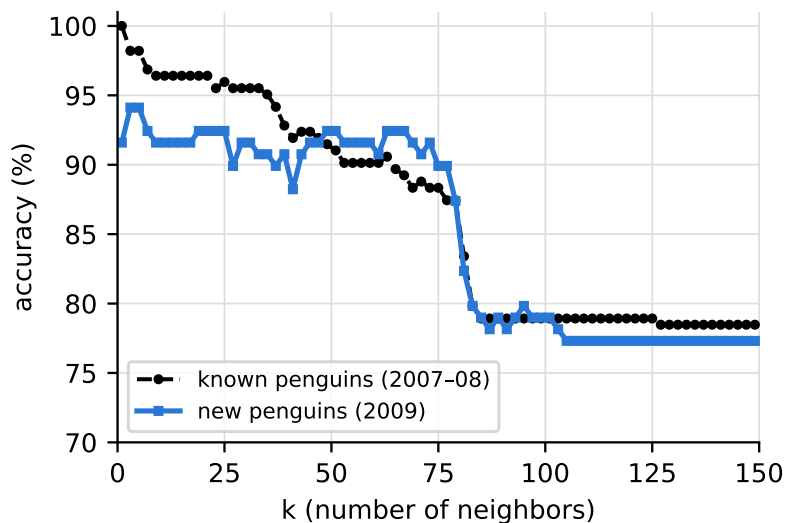


1. Zoomed in on **B**. One grid square is 1 mm × 1 mm, so distances are true.
2. **Circle** B's nearest neighbor. With $k = 1$, B is a _____.
3. Draw **one circle** around B's 5 nearest neighbors. Votes:
Adelie _____ Chinstrap _____ Gentoo _____
4. With $k = 5$, B is a _____.

3 · Three maps



1. The background shows what kNN predicts at every spot.
2. **Circle** one “island” in the $k = 1$ map.
3. At $k = 100$, **circle** what is left of the Chinstrap region.
4. Under each map, write *too wiggly*, *about right* or *too simple*.

4 · Which k ?

1. kNN's accuracy for every k : on the penguins it learned from, and on the new ones.
2. **Circle** the best k for the new penguins.
3. **Label** the overfitting zone and the underfitting zone.
4. At $k = 1$: known _____ %, new _____ %.